

10. Ethology (Gk. **ethos** - character)

The study of the animal behaviour based on the systematic observation, recording, analysis of functions of animals, with special attention to physiological, ecological and evolutionary aspects is called **ethology**.

1.4 NEED FOR CLASSIFICATION

It is impossible to study all living organisms. So, it is necessary to devise some means to make this possible. This process is called '**classification**'. Classification is defined as the process by which anything is grouped into convenient categories based on some easily observable characters. The scientific term used for these categories is 'TAXA' (singular: taxon). Taxa can indicate categories at different levels e.g. Animalia (which includes multicellular animals), chordata, mammalia, etc., represent taxa at different levels.

Hence, based on certain specific characteristics, all the living organisms can be classified into different taxa. This process of classification is called **taxonomy**. Study of external and internal structures, along with the structure of cell, developmental processes and ecological information of organisms are essential and they form the basis of modern taxonomic studies. Hence, **characterisation**, **identification**, **nomenclature** and **classification** are the processes that are basic to taxonomy.

'Zoos' As Tools For Classification

These are the places where wild animals, taken out of their natural habitat, are placed in protected environment under human care (**Ex-situ conservation**). This enables us to learn about the animal's external features, food habits, behaviour (ethology) etc. These observations enable us to systematise the organism and position it in the animal world.

Museums

Museums have BIOLOGICAL SPECIMENS preserved in containers or jars in preservative solutions. Animal specimens may also be preserved as dry specimens. Insects are preserved in insect boxes after collecting, killing and pinning on sheets. Larger animals like birds and mammals are usually 'stuffed' and preserved. Museums often have collections of skeletons of animals too.

1.5 BIOLOGICAL CLASSIFICATION

The living organisms exhibit a great deal of diversity due to variations in their structure and function. So far, over 1.25 million animal species have been identified and described. They show diversity in structure, habits, habitats and modes of life. To understand the interrelationships among the diversified animal groups, a systematic classification is necessary.